

ACT-SA-4000 Anchor Seal Assembly, 4.000 in. Bore

Technical Datasheet DS-SA-4000 Rev B | Product family: ACT-SA-4000

1. Product Overview

The ACT-SA-4000 anchor seal assembly latches into the 4.000 in. polished bore receptacle (PBR) of the ACT-LH-7625 liner hanger system, providing a pressure-tight, tension-holding connection between the production tubing and liner top. Molded bonded seal units ride in the honed PBR bore; a ratch-latch anchor resists tubing forces in both directions.

2. Applications

- Liner-top tieback seals with ACT-LH-7625 PBR
- Production string to liner isolation

3. Features

- Ratch-latch anchor - snaps in, releases with RH rotation (12 turns)
- 3 x bonded seal units, HNBR/PTFE (Aflas option for CO₂/H₂S)
- Locator sub allows space-out confirmation before latch
- Straight-bore molded seals rated 8,000 psi differential

4. Technical Specifications

Parameter	Specification
Mating PBR Bore	4.000 in. (ACT-LH-7625-08)
Seal OD (nominal)	3.995 in.
Minimum ID	2.441 in.
Number of Seal Units	3 (bonded, molded)
Pressure Rating	8,000 psi differential
Temperature Rating	325 degF (HNBR) / 375 degF (Aflas option)
Anchor Capacity	150,000 lbf tension
Release	12 turns RH rotation at latch
Connection (top)	2-7/8 in. EUE box (3-1/2 in. optional)

All ratings are at ambient temperature unless otherwise noted. Performance envelope per API 11D1 V3 qualification testing where indicated. Consult ACT Engineering for combined load conditions.

5. Materials of Construction

Component	Material
Mandrel / subs	AISI 4140 28-32 HRC (13Cr optional)
Seal units	Bonded HNBR 90A / PTFE anti-extrusion
Ratch-latch	AISI 4340 HT 36-40 HRC

6. Ordering & Reference Documents

Order by part number and specify: casing size and weight, service trim (standard or NACE sour), elastomer option, end connections, and any customer inspection requirements. Related documents:

- SOP-SA-4000 Spacing-Out Procedure
- DS-LH-7625
- TS-200

Datasheet DS-SA-4000 Rev B issued 2025-10-08. Prepared by R. Callahan; approved by J. Martinez, P.E.. Specifications subject to change; confirm with ACT Engineering for well-specific designs.